



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SUMI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

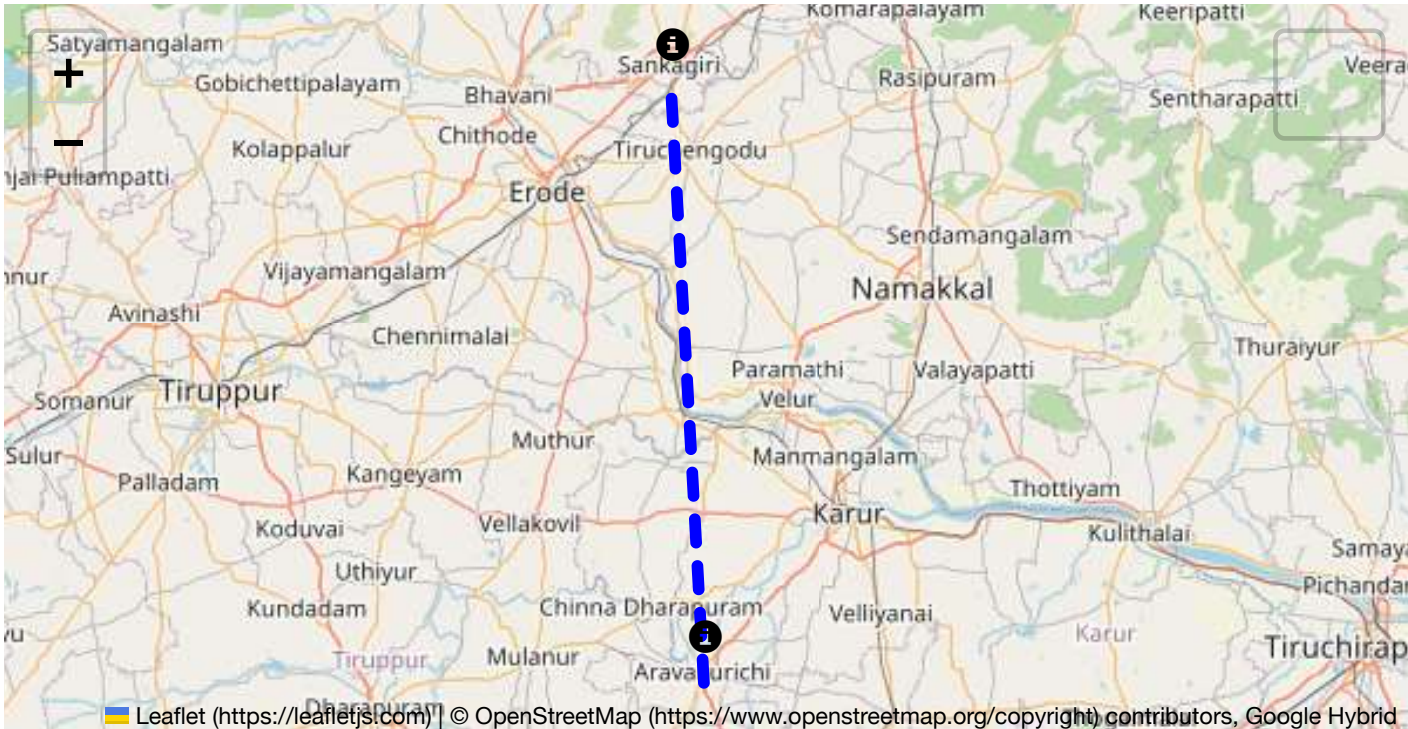
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

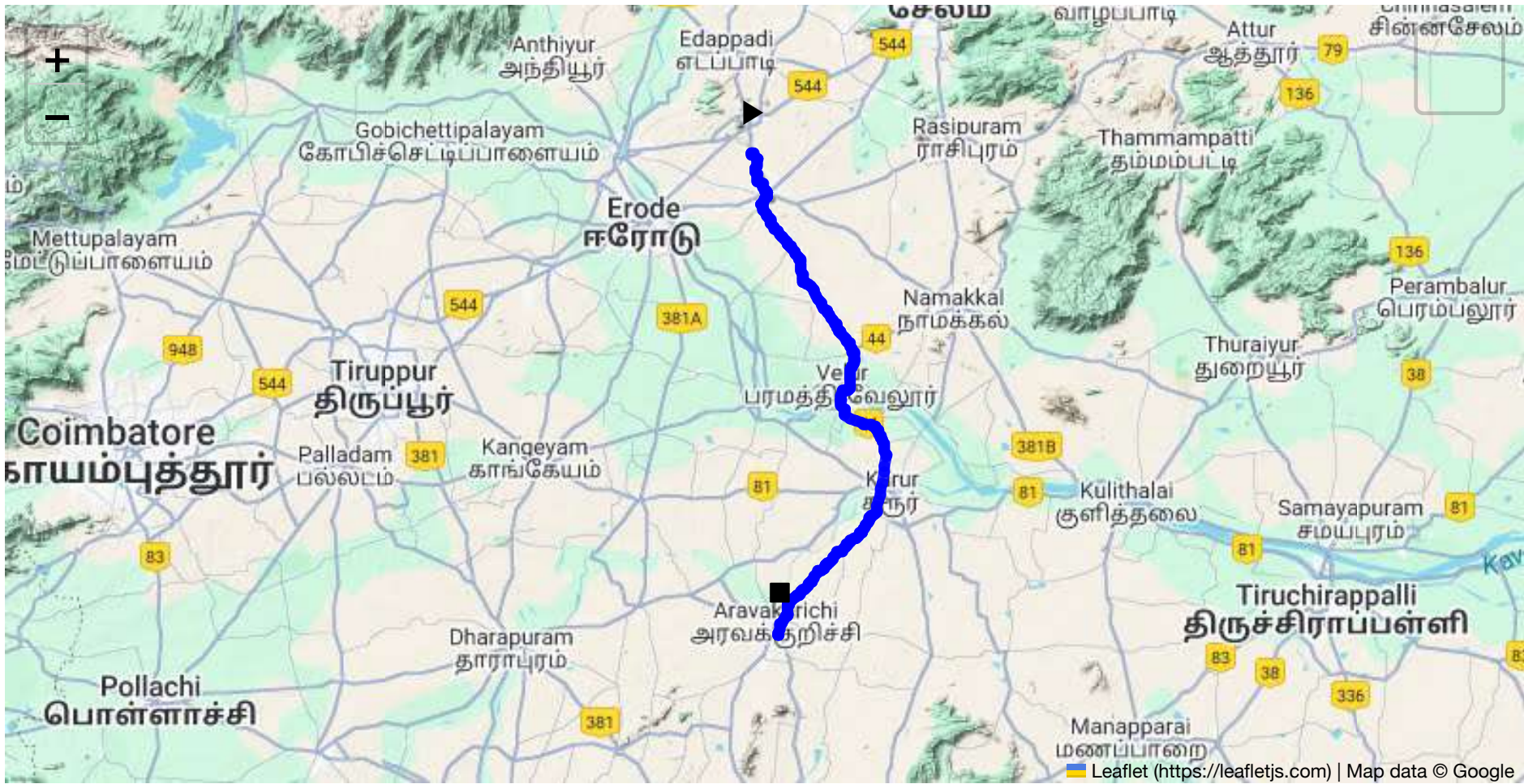
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 94.06 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours

Start: (11.4381, 77.8734)
End: (10.75517, 77.912405)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The journey from Sangagiri to NH 44 covers approximately 94.06 kilometers. It primarily involves travel on state highways and national highways, with the most significant portion of the distance covered via NH 44. The route is a mix of arterial roads that connect smaller towns and larger highway stretches.

2. Weather Conditions and Potential Weather-related Hazards

- Typical Weather:** The region experiences a tropical climate with hot summers, moderate monsoons, and mild winters.
- Potential Hazards:** During monsoon (June to September), heavy rain can lead to poor visibility, slippery roads, and possible flooding in low-lying areas. Fog may be an issue during early mornings in winter months.

3. Traffic Patterns

- Peak Hours:** Typically, congestion is highest during morning (7:30-9:30 AM) and evening (5:30-7:30 PM) rush hours, especially near urban centers.
- Congestion-prone Areas:** The towns along the route, particularly major intersections and near markets like Namakkal, can experience heavy traffic.

4. Road Quality and Infrastructure

- Road Quality:** Major highways like NH 44 are generally well-maintained with good surface quality, but secondary roads may exhibit potholes and uneven pavement.

- **Infrastructure:** Adequate signage and road markings are present on the highways, but rural and secondary roads may lack proper infrastructure.

5. Suggestions for Alternative Routes

In emergencies, an alternate route could involve traveling via SH 86 or SH 79 depending on the specific point of departure and incident-related obstructions.

6. Local Regulations for Hazardous Material Transport

- **Regulations:** Transport of hazardous materials requires following guidelines laid down by the Directorate General of Shipping in India, which includes proper labelling and adequate safety measures.
- **Curfews/Restrictions:** Movement of hazardous materials might be restricted during certain periods, especially during festivals or in congested areas.

7. Historical Incidents

There have been few recorded incidents involving heavy vehicles due to mechanical failure or driver fatigue. However, specific data on hazardous material incidents may require collaboration with local authorities to access.

8. Environmental Considerations

- **Sensitive Areas:** There are no major wildlife sanctuaries along this route, but attention should be given to agricultural zones to prevent contamination.
- **Waste Disposal:** Ensure no disposal of hazardous substances occurs along the route to prevent environmental damage.

9. Communication Coverage

Most of the route is covered by cellular networks, but there could be potential dead zones in rural stretches. Reliable communication is available near urban areas and along NH 44.

10. Estimated Emergency Response Times

- **Urban Areas:** Emergency response within 30-45 minutes due to proximity to services.
- **Rural Segments:** Response might take 60 minutes or more, depending on the specific location and availability of nearby emergency services.

12. Overall Summary of Risk Assessment

The route is generally safe for transporting hazardous materials, given compliance with regulations and careful monitoring of weather conditions. The chief risks include inclement weather during monsoon months, potential congestion in urban areas, and possible communication outages. Ensuring regular maintenance checks and adherence to transport guidelines can mitigate these risks effectively.

Emergency response availability is decent but consideration should be given to more rural stretches for planning breakdowns or incidents. Ensuring alternative route knowledge and readiness can further enhance driver safety and cargo security.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	Medium	11.15579, 78.02169	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14485, 78.02282	15 KM/Hr
11	Blind Spot	Blind Spot	10.75511, 77.91217	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
21	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15579, 78.02169



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14485, 78.02282



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.75511, 77.91217

